6.2.4 Valentine and Mesout (2009) — Eyewitness Identification Under Stress in the London Dungeon

What this workbook covers 本册讲什么

A real haunted attraction, real fear, and a nine-photo line-up. Visitors who felt **calm** identified the "scary person" **75%** of the time; those overwhelmed by anxiety managed only **17%**. This booklet covers both parts of the study — validating the anxiety questionnaire, then testing the witnesses — plus the evaluation and exam frames.

How to use this booklet 怎么用

1. **Fill it in as you go** — check the answer key (p. 20) after each page.
2. **Pages 5–10 (procedure & results) and 14 (numbers) matter most.** Do them from memory, under time.

Content map 内容地图

Page	Topic	What you must be able to do afterwards
p.2	Why the Dungeon 引入	Explain field setting + unaware witnesses + catastrophe model
p.3	Aim and two parts	State the hypothesis and why validation came first
p.4	Part one: validation	74.7 → 86.9 bpm · anxiety 43.2 correlated
p.5	State vs trait anxiety	Define and distinguish the two
p.6	Part two: sample	56 · 29 F / 27 M · mean 31.0 · reduced admission
p.7	Into the labyrinth	7 min · ~30 visitors · scary person · free vs cued recall
p.8	The line-up	Nine photos · culprit-present · "may or may not" · 0–100%
p.9–10	Results 结果	52.8/45.3 · 17% vs 75% · tables 12.5 & 12.6
p.11	Conclusions 结论	Anxiety impairs ID · females vulnerable · model supported
p.12–13	Evaluation 评价	Four strengths, three limitations with consequences
p.14	Every number 数字收口	Cover the right column and test yourself
p.15	Side by side 对照	vs Ruva et al. (2007)
p.16	Application 应用	Witness caution, admissibility, line-up limits
p.17	Exam skills 答题技巧	Command words, sentence frames
p.18–19	Practice & Final check	Short answers and a memory sweep
p.20	Answer key 答案速查	Mark honestly, then re-write wrong answers in full

Revision tracker 复习打卡

☐ p.2–3 setting & aim ☐ p.4–5 validation & anxiety types ☐ p.6–8 procedure ☐ p.9–11 results & conclusions ☐ p.12–13 evaluation ☐ p.14–19 exam

The whole study in one sentence 一句话总结

After showing the Dungeon genuinely raises heart rate (74.7 → 86.9 bpm), Valentine and Mesout had **56 visitors** meet a "scary person" in the Horror Labyrinth and then describe and identify them: only **17%** of witnesses with **high state anxiety** (above the median 52) made a correct identification, versus **75%** of calm witnesses — and males (19 accurate) beat females (7) — supporting the **catastrophe model**: high anxiety collapses eyewitness accuracy.

Why the London Dungeon? 为什么选鬼屋

The background 背景

Tim Valentine researched witness identifications for years. With Jan Mesout (2009) he wanted to **overcome limitations of laboratory research** on stress, by studying eyewitness identification in a **real-life scenario** — where witnesses were **not aware** their testimony would be tested.

Key terms 两个关键术语

The London Dungeon
A tourist attraction where visitors travel through rooms depicting London's history (e.g. the Great Fire), with **actors** acting out each theme.

catastrophe model of memory performance（突变模型）
_____ physiological arousal/anxiety can induce a _____ impairment of eyewitness recall and identification. (high · marked)

Fill in the blanks 填空

1. Lab participants may _____ the aims and respond how they think the researcher wants.

2. The Dungeon visitors did _____ know their testimony would be tested — like a genuine witness.

3. The study was designed to test the _____ model of memory performance.

「突变」是什么意思（中文）
不是慢慢变差，而是紧张超过某个程度后**断崖式崩盘**。数据先剧透：低焦虑 75% 指认成功，高焦虑只有 17%。

The Aim and the Two Parts 目的与两部分

Aim 研究目的

To test the hypothesis that _____ arousal (high levels of stress) can _____ the ability of an eyewitness to _____ information and _____ a perpetrator. (high · reduce · recall · identify)

Specifically: could visitors **describe and identify** the "scary person" from the **Horror Labyrinth**?

Two parts 两部分设计

Part	Purpose
1. Validation 验证	Show the Dungeon **really raises anxiety**, so the questionnaire is a _____ measure of stress (valid)
2. Eyewitness study 目击研究	Test _____ and _____ of the scary person against measured anxiety

为什么要先验证

如果被试只是「觉得累」或「觉得好玩」，问卷测的就不是焦虑。先用心率证明「鬼屋 = 真紧张」，问卷分数才有资格当自变量用。

Part One — Validating the Anxiety Measure 验证

The procedure 步骤

1. _____ employees of a London department store wore a **Polax Accurex wireless heart-rate monitor** strapped around the chest.
2. _____ heart rate recorded first, then monitored while walking around the Dungeons.
3. After the visit: completed a **State and Trait Anxiety Inventory**.
4. Data from _____ participants used (_____ had outlier baselines).

The numbers 结果

Measure	Value
Baseline heart rate	_____ bpm
Heart rate in the Horror Labyrinth	_____ bpm
Mean state anxiety score	_____
Relationship between the two	state anxiety _____ with heart-rate change → the inventory is a _____ measure

一句话总结验证部分

心率 74.7 → 86.9 bpm = 鬼屋真的吓人；焦虑分 43.2 与心率变化**相关** = 问卷测的确实是焦虑。

State vs Trait Anxiety 状态焦虑 vs 特质焦虑

Definitions 定义

state anxiety（状态焦虑）

Anxiety caused by _____ factors — how stressed you feel _____ in this situation.

trait anxiety（特质焦虑）

A measurable _____ difference / personality trait: a tendency to _____ about yourself and others _____ the actual circumstances. (individual · worry · regardless of)

Compare 对照

	State	Trait
Source 来源	The _____ (situational)	The _____ (dispositional)
Stable? 稳定吗	changes _____	_____ over time
Role in study 作用	split witnesses into high / low groups	_____ measure of individual differences

Spot the type 辨一辨

1. “My hands were shaking inside the maze.” → _____ anxiety

2. “I'm generally a worrier, even on a normal day.” → _____ anxiety

Fill the table 填表

Feature	Answer
Participants	_____ visitors to the London Dungeon
Sex	_____ female, _____ male
Mean age	_____ years
Incentive	a _____ in the admission price for completing questionnaires
When recruited	_____ the visit （after）

Ethics at this point 此处的伦理

After the tour, the _____ of the experiment was explained, _____ _____ was obtained, and participants could _____ at any stage.

评价伏笔

「事后招募 + 门票折扣」保住了**不知情**（生态效度），但制造了两个评价点：① **事先无法知情同意**；② 愿意停下来填问卷的人是**自选样本**。

The experience 体验

- Walked the **Horror Labyrinth** with around _____ other visitors, for about _____ minutes.
- A _____ designed to disorientate: _____ and scary _____ played on a recording.
- The “scary person”: an actor in a _____ robe, make-up to appear very _____, with _____ or _____ on the face.
- Behaviour: _____ out in front of the participant and _____ their way.
- All participants went around in the _____ direction.

The recall tests 回忆测验

Test	What it asked
Free recall 自由回忆	Describe the scary person _____ — measures how _____ you remember
Cued recall 线索回忆	Specific details: _____, age, height, hair colour — told **not to guess**
Questionnaires 问卷	_____ anxiety (in the dungeon) then _____ anxiety (general)

自由回忆 vs 线索回忆（中文）
自由回忆考**量**（记得多不多），线索回忆考**准**（细节对不对）。分开计分才知道焦虑伤的是哪一个。

The Line-up 指认

How it worked 流程

1. A _____-person photograph line-up; the scary person's photo placed in a _____ selected position.
2. Alongside _____ other similar-looking actors.
3. Instruction: “The person you saw _____ _____ _____ in the line-up.”
4. Guided to say if they could not _____ the person.
5. Then rated _____ in the decision from _____–_____ per cent.

Key term 关键术语

culprit-present line-up（目标在场指认列队）

A line-up that actually **contains** the person being searched for — so a wrong choice is a false identification, not a “target absent” catch.

为什么说「可能不在」（中文）

防止硬选：不说这句话，被试会默认「九张里必有一张」，乱猜也选一张。有了这句，「没认出」成为合法答案，指认成功才真实。—— 记住：后面应用题里这句指令**也救不了高焦虑证人**。

Results 1 — Anxiety Levels 焦虑结果

Fill in the numbers

Finding	Detail
State anxiety: females vs males	_____ vs _____ — females higher
Mean state anxiety (all)	_____
Mean trait anxiety	_____ — _____ difference between the sexes
Recall	Participants with _____ state anxiety recalled _____ correct information

Read the pattern 读规律

性别差异只在**状态焦虑**上（情境引起的），不在**特质焦虑**上（天生倾向）。这说明女性指认更差的原因是「当下更害怕」，不是「平时更爱操心」。

> **先立住「焦虑伤回忆」**
> 在看指认之前，回忆本身已经输了：低焦虑组回忆出的**正确信息更多**。这是结论的第一块证据。

Results 2 — The Identification Cliff 指认结果

The headline result 头条结果

The sample was split at the _____ state anxiety score (_____). Correct identification from the nine-person culprit-present line-up:

Group	Correct identification
LOW state anxiety (below the median)	_____%
HIGH state anxiety (above the median)	_____%

Table 12.5 全表（每组 28 人）

Identification	Low anxiety	High anxiety
Accurate	_____	_____
Inaccurate	_____	_____
No identification	_____	_____

Table 12.6 性别表

Identification	Male	Female
Accurate	_____	_____
Inaccurate	_____	_____
No identification	_____	_____

Two extras: correct identifiers showed **higher confidence** in their choice; and males made more correct identifications — the gender gap held even when state anxiety was accounted for separately.

Conclusions 结论

Five conclusions 五条结论

1. _____ show higher levels of anxiety in stressful situations.
2. Being highly anxious _____ the accuracy of eyewitnesses in identifying perpetrators.
3. Experts should take into account the _____ _____ of the witness at the time of the event.
4. Females may be particularly _____ to the effect of stress when witnessing or being a victim of crime.
5. The failure of high-anxiety witnesses supports the _____ _____ of memory performance under anxiety.

一句话版（中文）

紧张真的会砸了目击证词 —— 不是慢慢变差，是**断崖式崩盘**（75% → 17%）。司法系统评估证词时必须问：**案发时你有多害怕?**

Sentence building 造句

Write the conclusion sentence you would use in an exam: *“Valentine and Mesout concluded that…”*

Evaluation · Strengths 优点

Strength 1 — Genuine field experiment 真实现场

Natural setting; witnesses _____ they would be tested (like a genuine witness). More _____ than lab studies where participants may guess the aims. （not aware · ecological validity）

Strength 2 — Excellent controls → replicability 控制好可重复

_____ actor for every set of visitors, acting the same _____, on the same tour; every group experiences the same _____, in the same _____, for the same _____ of time. → high _____.

Strength 3 — Validated questionnaire 问卷验证过

Validated on a set of _____ _____ with heart-rate monitors → the inventory reliably measures _____, not another emotion.

Strength 4 — Individual differences 个体差异

Measuring _____ anxiety gave a _____ against which the state anxiety of the Dungeon could be compared — rather than just comparing a stress group with a control group.

> **评价句式（背结构）**
> **Point** → **Because/Example** → **Consequence**。四个优点各配一个后果。

Evaluation · Limitations 缺点

Limitation 1 — Self-selected scare-lovers 自选样本

Participants had all _____ to attend a scary place. They may _____ scary entertainment and be affected in a _____ way from people who avoid such places → limits _____ to the wider population.

Limitation 2 — Real fear before consent 事先无法同意

The fear was genuine and came _____ consent was possible (consent came only after the tour). Managed by debriefing and the right to _____, but the fear itself could not be consented to in advance.

Limitation 3 — Not a real crime 不是真案子

The scary person is an _____; the encounter lasts seconds inside a _____-minute maze with ~30 other visitors. A real crime may involve longer exposure, a weapon, or consequences — a different stress profile.

> **一体两面（高分思路）**
> 生态效度的**来源**（真实紧张、不知情）正是缺点的**来源**（无法事先同意、来的人爱找刺激）。优点缺点写成一对照应，是本题最漂亮的答案。

Cover the right column, test yourself 遮住右栏自测

Question	Answer
Year of the study	**2009**
Visitors in the eyewitness study (F / M)	**56** (29 / 27)
Mean age	**31.0**
Employees in validation (usable / outliers)	**20** (18 used / 2 outliers)
Baseline → Labyrinth heart rate	**74.7 → 86.9 bpm**
Mean state anxiety in validation	**43.2**
Labyrinth walk: companions and time	**~30 visitors, ~7 minutes**
Line-up size (composition)	**9** (culprit + 8 similar actors)
State anxiety — females / males	**52.8 / 45.3**
Mean state / trait anxiety (main study)	**49.0 / 36.8**
Median split for state anxiety	**52**
Correct ID — low / high anxiety	**75% / 17%**
Table 12.5 low / high (A / I / No-id)	low **21 / 6 / 1**; high **5 / 15 / 8**
Table 12.6 male / female accurate	**19 vs 7**
Confidence scale	**0–100%**

数字卡片

2009 · 56 (29/27) · 31.0 · 74.7→86.9 · 43.2 · 7 min · 9 photos · 52.8/45.3 · 49.0/36.8 · 75% vs 17% · 19 vs 7.

Complete the comparison 对照表（答案见 p.20）

Issue	Valentine and Mesout (2009)	Ruva, McEvoy and Bryant (2007)
Method	_____ experiment in a tourist attraction	Lab experiment with a real trial _____
Sample	_____ Dungeon visitors, mean age 31.0	_____ students, mean age 20.6
Focus	_____ on eyewitness identification	Pre-trial publicity and _____ memory errors
Key measures	State/trait anxiety, free vs cued recall, line-up ID	Verdicts, guilt ratings, _____ _____ test
Headline result	_____% vs 17% correct identification	73% vs 41% _____ verdicts
Deception?	No cover story, but no _____ consent before the fear	_____ story used; most believed it
Shared theme	Both test how a _____-courtroom factor (stress / media) distorts eyewitness or juror decisions	

怎么背这张表

一句话反差：**V&M 测「当下的恐惧」对目击指认的影响；Ruva 测「过去的新闻」对裁决的影响**。共同点：法庭之外的因素会扭曲司法判断。

Application — Real-life Implications 应用

Three implications 三条应用

1. During trials of real-life crimes, be _____ in trusting accounts from witnesses who underwent significant _____ or _____ during the crime.
2. Police should be informed that anxiety can _____ recall and identification — a consideration when assessing the _____ of eyewitness testimony for a trial.
3. Also a consideration when police use _____ identification techniques.

The subtle point 细节考点

Valentine and Mesout told participants the person _____ _____ _____ — yet anxious witnesses still failed. So this instruction alone is _____ to ensure witnesses do not select the wrong suspect.

> **这条应用为什么高分（中文）**
> 不止说「小心紧张证人」，还顺着研究设计往前多走一步：**标准指令挡不住焦虑对记忆的破坏** —— 因为破坏发生在记忆形成时，不在选择时。

Command words 指令词

Command word	What it demands
Describe	Facts only — two parts, sample, labyrinth, tests, line-up.
Explain	The **because/why** — e.g. why the questionnaire had to be validated first.
Evaluate / assess	Strength + limitation, each with a consequence, then a verdict.

Sentence frames 万能句架

- Procedure: *"56 visitors walked the Horror Labyrinth for around seven minutes, where an actor — the 'scary person' — stepped out and blocked their path…"*
- Result with numbers: *"Only 17% of high-anxiety witnesses correctly identified the scary person, compared with 75% of low-anxiety witnesses."*
- Model sentence: *"The failure of highly anxious witnesses to identify the perpetrator supports the catastrophe model of memory performance."*
- Evaluation: *"One strength is that witnesses were unaware they would be tested, which means… However, participants had all chosen to visit a scary attraction, so…"*

四个必写细节

① **两部分设计**（先验证后实验）② **75% vs 17%**（头条数字）③ **catastrophe model**（模型名）④ **unaware witnesses**（生态效度来源）。

Practice A — Short Answers

Q1. Describe the sample used in Valentine and Mesout (2009). (4)

Q2. Explain why the researchers first validated the State Anxiety Inventory. (4)

Q3. Outline how the line-up identification was carried out. (4)

Q4. Describe the difference between state anxiety and trait anxiety, with one example of each from the study. (4)

Final Check — Five Minutes, No Notes

Ten answers from memory 十题默检

1. Year and setting: _____
2. Sample size and sex split: _____
3. Heart rate before / in the labyrinth: _____
4. What the scary person wore and looked like: _____
5. Length of the labyrinth walk (and companions): _____
6. Median state anxiety score for the split: _____
7. Correct identification — low vs high anxiety: _____
8. Which sex made more correct identifications: _____
9. Name of the model supported: _____
0. One application for police or courts: _____

Brain dump 空白默写

Close the booklet. Write everything you remember about Valentine and Mesout (2009) — aim → sample → procedure → one result → one evaluation point.

Answer Key 答案速查

Page 2

1. guess 2. not 3. catastrophe. Box: high · marked.

Page 3

high · reduce · recall · identify · valid · recall · identification

Page 4

20 · Initial · 18; 2 · 74.7 · 86.9 · 43.2 · correlated; valid (reliable)

Page 5

situational; now · individual; worry; regardless of · situation; person · moment to moment; consistent · baseline · 1. state 2. trait

Page 6

56 · 29; 27 · 31.0 · reduction · after · purpose; informed consent; withdraw

Page 7

30; 7 · maze; dark; sounds · dark; pale; wounds; scars · step; blocked · same · own words; much · gender; state; trait

Page 8

nine; randomly · eight · may or may not be · identify · confidence; 0–100

Page 9

52.8; 45.3 · 49.0 · 36.8; no · lower; more

Page 10

median; 52 · 75%; 17% · Table 12.5: 21/6/1; 5/15/8 · Table 12.6: 19/5/5 male; 7/16/4 female

Page 11

Females · reduces · emotional state · vulnerable · catastrophe model

Pages 12–13

p.12: not aware; ecological validity · same; manner; exhibits; order; period; replicability · office workers; anxiety · trait; baseline. p.13: chosen; prefer; different; generalisability · before; withdraw · actor; 7.

Page 15

field; video · 56; 558 · anxiety; source · source monitoring · 75; guilty · prior; cover · outside

Page 16

cautious; anxiety; trauma · impair; admissibility · line-up · may or may not be present; not enough

Page 18 — Practice A (model points)

Q1 56 visitors to the London Dungeon (29 female, 27 male), mean age 31.0, recruited after their visit with a reduction in admission price for completing questionnaires.
Q2 To show the Dungeon genuinely raises anxiety: 20 employees' heart rate rose 74.7 → 86.9 bpm in the labyrinth, and mean state anxiety (43.2) correlated with the heart-rate change — so the inventory validly measures stress rather than another emotion.
Q3 Nine-person culprit-present photo line-up; scary person's photo in a random position with eight similar actors; told the person may or may not be present; could say if unable to identify; confidence rated 0–100%.
Q4 State = situational, felt now (terror in the dark maze); trait = stable disposition to worry regardless of circumstances (measured as a personal baseline). Females showed higher state (52.8 vs 45.3) but equal trait anxiety.

答错怎么办

把答错的题**整句重写一遍**（不是只补答案），隔天再默一次。错两次以上的点，誊到 p.19 brain dump 区。